

A Primitive Polynomial to Define the Finite Field Modules for High-Throughput Interpolator to Encode and Decode Of Non Binary Cyclic Codes

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Abstract : This paper represents The algebraic soft-decoding (ASD) of Reed–Solomon (RS) codes provides significant coding gain over hard-decision decoding with polynomial complexity. The low-complexity chase (LCC) algorithm is proposed for reducing the complexity of interpolation, which interpolates over 2^n test vectors, being attractive for VLSI implementation. The interpolation is simplified in LCC decoding by restricting the multiplicity to $m=1$ and replacing the factorization step with Chien's search and Forney's algorithm. In this paper, high-throughput interpolator architecture for soft-decision decoding of Reed–Solomon (RS) codes based on low-complexity chase (LCC) decoding is presented. We have formulated a modified form of the Nielson's interpolation algorithm, using some typical features of LCC decoding. The proposed algorithm works with a different scheduling, takes care of the limited growth of the polynomials, and shares the common interpolation points, for reducing the latency of interpolation. Based on the proposed modified Nielson's algorithm we have derived low-latency architecture to reduce the overall latency of the whole LCC decoder. An efficiency is low, in terms of area-delay product, has been achieved by an LCC decoder, by using the proposed interpolator architecture, over the best of the previously reported architectures for an RS(255,239) code with eight test vectors.

Keyword: RScodes, Guruswami-Sudan algorithm, Registers, Multiplexers, D-flipflop.

I. INTRODUCTION

2. Interpolation

3. Factorization of bivariate polynomials

It's being the interpolation and the most computation-intensive one. Several architectures based on Nielson's algorithm and Lee–O'Sullivan algorithm are found in the literature for the VLSI implementation of interpolation stage. However, their hardware complexity is still high. An interpolation architecture for LCC with $m=3$, called backward interpolation, is proposed in [9], which could be considered as the best of the current approaches. Backward interpolator shares the computation of common points of the test vectors. These points are ordered in such way that a pair of adjacent vectors differ only at one point.

Due to this feature, the backward interpolation architecture involves less area and provides higher speed than its prior ones.

II. REED SOLOMON SOFTWARE PERFORMANCE

The following data details performance numbers for a number of specific RS (n, k) implementations for two general purpose processing architectures and one digital signal processor. Numbers are provided for both decode in the presence of no error, as well as decode in the presence of maximum channel error. Note that correcting errors requires more processing power than simply validating blocks, and that the required processing power increases linearly with the error rate. Typical applications tend to keep the error rate low such that active correction is not required. The two digit hexadecimal number in each column specifies the GF (255) primitive polynomial used to generate the underlying Galois field. In 1960, Irving Reed and Gus Solomon published a paper in the Journal of the Society for Industrial and Applied Mathematics. This paper described a new class of error-correcting codes that are now called Reed-Solomon (R-S) codes. These codes have great power and utility, and are today found in many applications from compact disc players to deep-space applications. This article is an attempt to describe the paramount features of R-S codes and the fundamentals of how they work. Reed-Solomon codes are non-binary cyclic codes with symbols made up of m-bit Sequences, where m is any positive integer having a value greater than 2. R-S (n, k) codes on m-bit symbols exist for all n and k for which $0 < k < n < 2m + 2$

Where k is the number of data symbols being encoded, and n is the total number of code symbols in the encoded block. For the most conventional R-S (n, k) code, $(n, k) = (2m - 1, 2m - 1 - 2t)$

Where t is the symbol-error correcting capability of the code, and $n - k = 2t$ is the number of parity symbols. An extended R-S code can be made up with $n = 2m$ or $n = 2m + 1$, but not any further. Reed-Solomon codes achieve the largest possible code minimum distance for any linear code with the same encoder input and output block lengths. For non-binary codes, the distance between two code words is defined (analogous to Hamming distance) as the number

Design of Modules to Implement a Structure by Discrete Reckoning Codes for Embedding Into Video Coding Testing Applications

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Abstract : In this paper, necessary modules have been implemented to make architecture to insert into motion estimation. The significant function of motion estimation (ME) in a video coder, testing such a module is of priority concern. While focusing on the testing of ME in a video coding system, this work presents an error detection and data recovery (EDDR) design, based on the residue-and quotient (RQ) code, to embed into ME for video coding testing applications. An error in processing elements (PEs), i.e. key components of a ME, can be detected and recovered effectively by using the proposed EDDR design. Experimental results indicate that the modules required for proposed EDDR design for ME testing can detect errors and recover data with an acceptable area overhead and timing penalty. Importantly, the proposed EDDR design performs satisfactorily in terms of throughput and reliability for ME testing applications.

Keywords – Motion Estimation, SAD Tree, Error detection and recover circuits.

I. INTRODUCTION

Methodologies

Coding approaches such as parity code, Berger code, and residue code have been considered for design applications to detect circuit errors. Residue code is generally separable arithmetic codes by estimating a residue for data and appending it to data. Error detection logic for operations is typically derived by a separate residue code, making the detection logic is simple and easily implemented. For instance, assume that N denotes an integer, N1 and N2 represent data words, and m refers to the modulus. A separate residue code of interest is one in which N is coded as a pair. $N \cdot m$ is the residue N of m modulo . Error detection logic for operations is typically derived using a separate residue code such that detection logic is simply and easily implemented. However, only a bit error can be detected based on the residue code. Additionally, an error can not be recovered effectively by using the residue codes. Therefore, this work presents a quotient code, which is derived from the residue code, to assist the residue code in detecting multiple errors and recovering errors. In order to simplify the complexity of circuit design, the implementation of the module is generally dependent on the addition operation. Additionally, based on the concept of residue code, the following definitions shown can be applied to generate the RQ code for circuit design. the corresponding circuit design of the RQCG is easily realized by using the simple adders (ADDs). Namely, the

RQ code can be generated with a low complexity and little hardware cost.

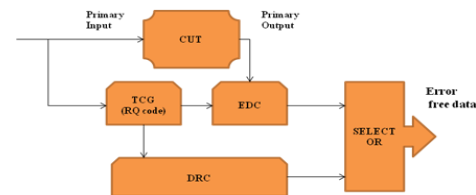


Fig 1: Circuit Diagram

Sum Of Absolute Difference Calculation

By utilizing PEs, SAD shown in as follows, in a macro block with size N X N of can be evaluated:

$$SAD = \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} |X_{ij} - Y_{ij}|$$

$$= \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} |(q_{xij} \cdot m + r_{xij}) - (q_{yij} \cdot m + r_{yij})|$$

Where $r_{xi,j}$, $q_{xi,j}$ and $r_{yi,j}$, $q_{yi,j}$ denote the corresponding RQ code of $X_{i,j}$, $Y_{i,j}$ and modulo M. Importantly, and represent the luminance pixel value of Cur_pixel and Ref_pixel, respectively.

PIXEL VALUES

	0	1	2	3
0	128	128	64	255
1	128	64	255	64
2	64	255	64	128
3	255	64	128	128

Cur_pixel

	0	1	2	3
0	1	1	2	3
1	1	2	3	4
2	2	3	4	5
3	3	4	5	5

Ref_pixel

Fig 2: Circuit Diagram

NUMERIC CALCULATION

A numerical example of the 16 pixels for a 4X 4 macroblock in a specific PEi of a ME is described . We presents an example of pixel values of the Cur_pixel and Ref_pixel. Based on, the SAD value of the 4X 4 macroblock is

$$SAD = \sum_{i=0}^3 \sum_{j=0}^3 |X_{ij} - Y_{ij}|$$

$$= X_{00} - Y_{00} + |X_{01} - Y_{01}| + \dots + X_{33} - Y_{33}$$

$$= (128 - 1) + (128 - 1) + \dots + (128 - 5)$$

$$= 2124.$$

According to describe about RQ Code for modulo operation is assumed $M=2^6-1=63$ RQ code for SAD value $RT=RPEi=|2124|63=45$ and $QT=QPEi=|2124/63|=33$.

JPEG IMAGE COMPRESSION USING VERILOG

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Abstract

Data compression is the reduction or elimination of redundancy in data representation in order to achieve savings in storage and communication costs. Data compression techniques can be broadly classified into two categories: Lossless, Lossy schemes. Digital images require an enormous amount of space for storage. The architecture exploits the principles of pipelining and parallelism to the maximum extent in order to obtain high speed the architecture for discrete cosine transforms and the entropy encoder are based on efficient algorithms designed for high speed VLSI. For example, a colour image with a resolution of 1024 x 1024 picture elements (pixels) with 24 bits per pixel would require 3.15 Mbytes in uncompressed form. Very high-speed design of efficient compression techniques will significantly help in meeting that challenge. The JPEG baseline algorithm consists mainly of two parts: (i) Discrete Cosine Transform (DCT) computation and (ii) Entropy encoding. The architecture for entropy encoding is based on a hardware algorithm designed to yield maximum clock speed

Index Terms: Data Compression, Discrete Cosine Transform, Entropy Encoding, JPEG.

1. INTRODUCTION

Today we are talking about digital networks, digital representation of images, movies, video, TV, voice, digital library-all because digital representation of the signal is more robust than the Analog counterpart for processing, manipulation, storage, recovery, and transmission over long distances, even across the globe through communication networks. In recent years, there have been significant advancements in processing of still image, video, graphics, speech, and audio signals through digital computers in order to accomplish different application challenges. As a result, multimedia information comprising image, video, audio, speech, text, and other data types has the potential to become just another data type[1][2][3]. Still image and video data comprise a significant portion of the multimedia data and they occupy the lion's share of the communication bandwidth for multimedia communication. As a result, development of efficient image compression techniques continues to be an important challenge to us, both in academia and in industry.

In an abstract sense, we can describe “data compression” as a method that takes an input data D and generates a shorter representation of the data $c(D)$ with a fewer number of bits compared to that of D . The reverse process is called “decompression”, which takes the compressed data $c(D)$ and generates or reconstructs the data D' as shown in Figure 1.1. Sometimes the compression (coding) and decompression (decoding) systems together are called a “CODEC,” as shown in the broken box in Figure 1.1.

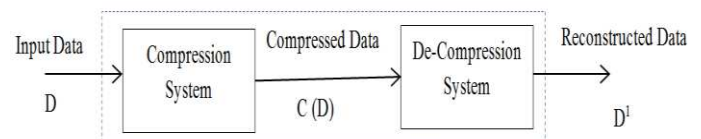


Fig-1: Codec

2. PREVIOUS WORK

Usually we need to apply lossless data compression techniques on text data or scientific data. For example, we cannot afford

Synthesis of a MSP430 Microcontroller Core using Multi-Voltage Techniques

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Abstract— This paper describes regarding Implementation of Multi-Voltage low power techniques for power improvement. Here Multi-Voltage technique is applied to MSP430 16-bit microcontroller core victimization TSMC 65nm & 45nm NLDM libraries. The optimized power values area unit shown and compared among the libraries. Synopsys Power Compiler tool is employed for synthesis and implementing the Technique. Final outline and Conclusion provided is predicated on the results obtained much once implementation. Reports generated once synthesis state that there is 96.4% reduction in escape power and 26.7% reduction in dynamic power for 65nm NLDM library and 94.4% reduction in escape power and 37.8% reduction in dynamic power for 45nm NLDM library.

Keywords— Multi-Voltage technique, TSMC, leakage power, MSP430.

I. INTRODUCTION

With the growing popularity of portable devices, the consumers need for electronics has been focusing on delivering the right performance with right power with minimal power waste[1]. On the one hand, product specifications dictate their designs must consume minimum amounts of power to meet cost, reliability, and energy efficiency goals. Secondly, functional specifications imply that the ASICs must integrate more circuitry and run faster than ever to implement the desired functionality, which in turn drives up overall power consumption[2].

As a result, a way to minimize power consumption whereas delivering the required performance and conjointly to extend battery period has become a significant concern and a crucial style goal that trendy chip designers should lookout.

The total power for a design consists of dynamic power and static power.

A. Dynamic Power

Dynamic power is that the energy consumed throughout logic transitions on nets, consisting of 2 elements, shift power and internal power. shift power results from the charging and discharging of the external electrical phenomenon load on the output of a cell. Internal power results from the short-circuit (crowbar) current that flows through the PMOS NMOS stack throughout a transition

The cause of switching power is illustrated in Figure 1

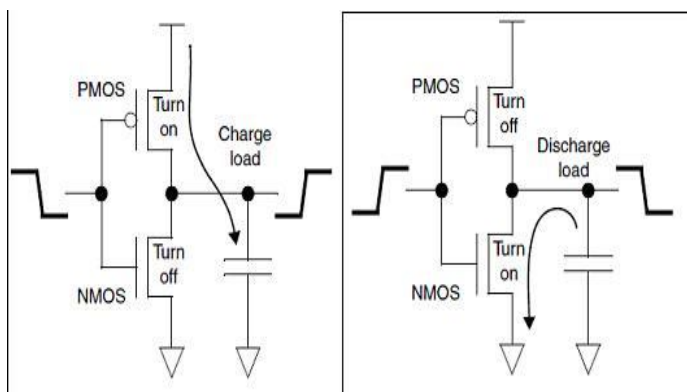


Fig. 1 Switching Power [2]

Simulation & Synthesis of a Cryptography Processor for Portable Electronic Devices

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Abstract — Cryptography circuits for portable electronic devices provide user authentication and secure data communication. These circuits should, in general, occupy small chip area, consume low power, handle several cryptography algorithms, and provide acceptable performance. This paper presents the simulation and synthesis of three standard cryptography algorithms on a universal architecture. The cryptography processor implements both private key and public key algorithms and meets the power and performance specifications. The mentor graphics modelsim tool is used for design and simulation and also Synopsys Design Compiler tool is used for synthesis. TSMC 65nm library is used for the synthesis.

Keywords — Synthesis, Cryptography Processor, AES, DES, ECC.

I. INTRODUCTION

The rapid growth of portable electronic devices with limited power and area has opened a vast area of low-power and compact circuit design opportunities and challenges for VLSI circuit designers. Cellular phones, PDAs, and smart cards are examples of portable electronic products that are becoming an integral part of everyday life. The popularity of these devices necessitates special considerations for their security subsystems. Unlike computer network security systems that impose less stringent limitations on the area and power consumption but put more emphasis on high throughput (several Gigabit/s), portable applications demand security hardware with more restrictions on area and power and less on throughput (several hundred kilobit/s to a few Megabit/s). This difference in requirements dictates a different approach in the design and implementation of the security systems for these devices.

Since next-generation, portable electronic devices will be used for a wide range of applications their security system must implement both private (symmetric) and public (asymmetric) key algorithms, to accommodate various application requirements.

Private Key algorithms with high throughput are suitable for data communication, while public key algorithms with much lower throughput are suitable for private key exchange and authentication. Among all available algorithms, data encryption standard (DES), advanced encryption standard (AES), and elliptic curve

cryptography (ECC), which are approved by standards organizations [1]–[3], are selected for this application. DES, for past compatibility, and AES, for high security and throughput, are the major candidates for private key algorithms, and ECC is the best candidate for the public key algorithm for its encryption efficiency.

A cryptography system can be implemented in either software or hardware. Software implementations allow multiple algorithms to be supported on the same hardware platform, but they are usually slow and cannot meet the required specifications. Moreover, they are considered to be more vulnerable to side-channel attacks compared to other implementations. Side-channel attacks use physical measurements on the device, for example, the power consumption of the processor, to detect the encryption/decryption key [4]–[6].

This paper introduces, a synthesis implementation of a cryptography coprocessor (crypto-processor) that implements two standard private-key algorithms (DES and AES), and one standard public-key algorithm (ECC), in a single design

II. PROBLEM STATEMENT

In this section, a brief introduction of the three implemented Algorithms is presented.

A. Data Encryption Standard (DES)

This is a well-established algorithm that has been used for more than two decades (since 1977) in military and commercial data exchange and storage. The algorithm is designed to encipher and decipher blocks of data consisting of 64 b using a 56-b key. A block to be enciphered is subjected to an initial permutation (IP), then to 16 rounds of a complex key-dependent permutation, and, finally, to another permutation which is the inverse of the IP, as shown in Fig. 1. The function $f()$ in this figure is the heart of this algorithm and consists of an expansion, XOR, lookup table (LUT), and permutation, as depicted in Fig. 2. To decipher, it is necessary to apply the very same algorithm to an enciphered message block, using the same key.



AN FPGA BASED 64-BIT IEEE – 754 DOUBLE PRECISION FLOATING POINT ADDER/SUBTRACTOR AND MULTIPLIER USING VHDL

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Abstract:

Floating point arithmetic is widely used in many areas, especially scientific computation and signal processing. For many signal processing, and graphics applications, it is acceptable to trade off some accuracy (in the least significant bit positions) for faster and better implementations. In this paper we describe an implementation of high speed IEEE 754 double precision floating point multiplier targeted for Xilinx Virtex-6 FPGA. VHDL is used to implement the design. Floating Point (FP) addition, subtraction and multiplication are more used in large set of scientific and signal processing computation. A high speed floating point double precision adder/subtractor and multiplier are implemented on a Virtex -6 FPGA. In addition, the proposed design is compliant with IEEE-754 format and handles overflow, under flow, rounding and various exception conditions.

Key Words: Double Precision, Floating Point, Adder/Subtractor, Multiplier, FPGA, IEEE 754 & Virtex-6.

Introduction:

Programming languages support numbers with fraction Called floating-point numbers. Scientific notation: $\pm d.f_1f_2f_3f_4 \dots \times 10^{\pm e_1e_2e_3}$

Examples:

3.14159265... (π)

2.71828... (e)

Floating Point Arithmetic is widely used in many areas, especially scientific computation and signal processing. The advantage of floating-point representation over fixed-point and integer representation is that it can support a much wider range of values. Floating point numbers are one possible way of representing real numbers in binary format; the IEEE 754 standard presents two different floating point formats, Binary interchange format and Decimal interchange format. Defined by IEEE Std 754-1985, floating points have 2 representations.

Sign bit (1 bit)	Exponent bit (8 bits)	Mantissa (23 bits)
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Two representations

- Single precision (32-bit)

1-bit sign + 8-bit exponent + 23-bit fraction

IEEE 754 uses biased representation for the exponent

- Value of exponent = $\text{val}(E) = E - \text{Bias}$.
- exponent field is 8 bits for single precision E can be in the range 0 to 255

ADAPTIVE HEARING AID ALGORITHM USING DIFFERENT TYPES OF MULTIPLIER

M.Aravindkumar, P.Sivananthamaitrey, K. RameshChandra • Published 2016 • Engineering, Computer Science

TLDR Flexibility of gain processing, updating filter coefficients using adaptive techniques and digital feed back reduction, etc, and major focus in this paper is given to the impact of multipliers on the power consumption of digital hearing aids.[Expand](#)

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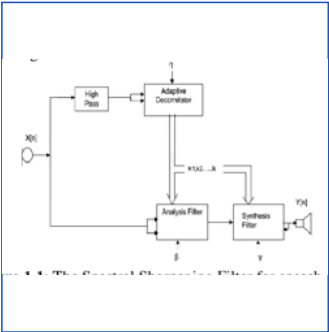


Figure 1.1

TABLE 2: TABLE 2: algorithm

S.NO	ms+1	ms	ms-1	Reoded digit	Operation on the multiplicand
1	0	0	0	0	0 X multiplicand
2	0	0	1	+1	+1 X multiplicand
3	0	1	0	+1	+1 X multiplicand
4	0	1	1	+2	+2 X multiplicand
5	1	0	0	-2	-2 X multiplicand
6	1	0	1	-1	-1 X multiplicand
7	1	1	0	-1	-1 X multiplicand
8	1	1	1	0	0 X multiplicand

Table 2

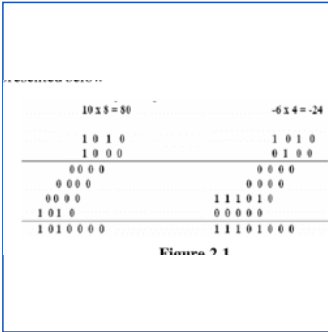


Figure 2.1

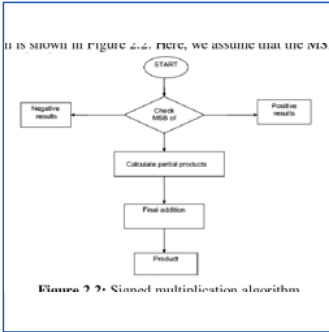


Figure 2.2

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Topics

AI-Generated

- Booth Multipliers
- Spectral Sharpening
- Multiplier
- Adaptive Filters
- Field Programmable Gate Array

Performance Evaluation of Different Multipliers in VLSI using VHDL

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Abstract: Multiplier modules are common to many DSP applications. The fastest types of multipliers are parallel multipliers. Among these, the Array multiplier is the basic one. However, they suffer from more propagation delay. Hence, where regularity, high performance and low power are primary concerns, Booth multipliers tend to be the primary choice. Booth multipliers allow the operation on signed operands in 2's-complement which are derived from array multipliers where each bit in a partial product line an encoding scheme is used to determine whether the bit is positive, negative or zero. The Modified Booth algorithm achieves a major performance improvement through radix-4 encoding. In this algorithm each partial product line operates on 2 bits at a time, thereby reducing the total number of the partial products. This is particularly true for operands using 16 bits or more.

Keywords: Array multiplier, parallel multiplier, propagation delay, VHDL, LUT, DSP block, utilization and twiddle factors.

1. INTRODUCTION

Performance is recognized as a one of the critical parameter in digital system design field, especially in digital signal processing (DSP) applications. Fast multipliers are essential parts of digital signal processing systems. The speed of multiply operation is of great importance in DSP as well as in the general purpose processors today, especially since the media processing took off. In the past multiplication was generally implemented via a sequence of addition, Subtraction, and shift operations. Multiplication operation is the series of repeated additions, number to be added is the multiplicand and the number of times that it is added is the multiplier, and the result is the product. Each step of addition generates a partial product. In most computers, the operand usually contains the same number of bits. When the operands are interpreted as integers, the product is generally twice the length is determined by the performance of the multiplier because the multiplier is generally the slowest element in the system. In digit, serial multipliers have single digits consisting of several bits are operated on. These multipliers have moderate performance in both speed and area. However, existing digit serial multipliers have been plagued by complicated switching systems or irregularities in design. The parallel Multipliers provide high performance compared to serial multipliers and avoiding irregularities. In this paper, evaluating performance of 3 different parallel multipliers in terms of power, delay and area and find out which one is providing high performance of operands in order to preserve the information content. This repeated addition method that is suggested by the arithmetic definition is slow that it is almost always replaced by an algorithm that makes use of positional representation. It is possible to decompose multipliers into two parts. The first part is dedicated to the generation of partial products, and the second one collects and adds them. The basic multiplication principle is

twofold, evaluation of partial products and accumulation of the shifted partial products.

It is performed by the successive Addition's of the columns of the shifted bit of the 'multiplicand'. The delayed, gated instance of the multiplicand must all be in the same column of the shifted partial product matrix. They are then added to form the product bit for the particular form .Multiplication is there for multi operand operation. To extend the multiplication to both signed and unsigned numbers a convenient. Number system would be the representation of numbers in two's complement format. Multiplication is an important fundamental function in arithmetic function.

2. MULTIPLIERS

Multipliers play an important role in today's digital signal processing and various other applications. With advances in technology, many researchers have tried and are trying to design multipliers which offer either of the following design targets – high speed, low power consumption, regularity of layout and hence less area or even combination of them in one multiplier thus making them suitable for various high speed, low power and compact VLSI implementation.

Characteristics of multiplier:

An efficient multiplier should have following characteristics:

Speed: Multiplier should perform operation at high speed.

Area: A multiplier should occupy less number of slices and LUTs.

Power: Multiplier should consume less power.

Multiplication process has three main steps:

1. Partial product generation.
2. Partial product reduction.
3. Final addition.

AN IOT-BASED EFFICIENT WEARABLE FALL DETECTION SYSTEM USING RASPBERRY-PI

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ABSTRACT: Falling is the most hazardous occasions that frequently occur among senior individuals, patients which requires medicinal consideration in time. Fall discovery frameworks could support senior individuals and patients to live autonomously. A programmed continuous fall discovery framework may discover fall occasions among old individuals in time that lessen the general setback rate. Fall detection (FD) and fall prevention (FP) are research areas that have been active for over a decade, and they both strive for improving people's lives through the use of pervasive computing. This paper surveys the state of the art in FD and FP systems, including qualitative comparisons among various studies. Proposed system used to find patient fall prediction and used to monitor patient body parameter conditions like pressure, temperature.

KEYWORDS: *Fall detection (FD), Fall prevention (FP), Raspberry-Pi, Internet of Things, Accelerometer*

INTRODUCTION: Normally it is difficult to keep track on abnormalities in heartbeat count for patient itself manually. The average heartbeat per minute for 25-year old ranges between 60- 80 beats per minute while for a 60-year old it is typically between 50-60 beats per minute and body temperature is 37degree Celsius or 98.6 Fahrenheit. Patients are not well versed with manual treatment which doctors normally use for tracking the count of heartbeat. So there must be some device which would help patient to keep track on their health by themselves. There are various instruments available in market to keep track on internal body changes. But there are many limitations regarding their maintenance due their heavy cost, size of instruments, and mobility of patients. To overcome these limitations a device use to keep track on heartbeat count of patient should be easy to use, portable, light weighted, small size etc so that it give freedom of mobility for patient. The devices which can be carried everywhere to keep track on patient's health. This device that is a heartbeat sensor would help them to keep track on heartbeat counts of a patient and check for any abnormalities. If any varied change takes place it is notified. This notification would help to take an appropriate action at an instance of a time. This would save patients from the future health problem which would arise. This would also help patient's concern doctor to take an appropriate action at proper time. The process of implementing a damage detection and characterization strategy for engineering structures is referred to as Structural Health Monitoring (SHM). Here damage is defined as changes to the material and/or geometric properties of a structural system, including changes to the boundary conditions and system connectivity, which adversely affect the system's performance. The

Early Detection of Diabetic Foot Ulcer by Using IoT Technology

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Periodicity: August - October'2017

DOI : <https://doi.org/10.26634/jfet.13.1.13762>



1

Total citation



0

Recent citations



n/a

Field Citation Ratio



n/a

Relative Citation Ratio

Abstract

Diabetic patients generally develop the complications of foot ulcers, breaking down of the skin tissue in the case poorly controlled diabetics. As the numbers of diabetic patient are increasing globally with the change in the lifestyle, the measures for the early detection of the diabetic related problems are gaining medical importance. Diabetic patients develop foot ulcers because of improper blood flow, loss of the sensation, and low wound healing rate. The proposed paper presents a system, which will provide readings of the hardness level at various parts of the foot. This is done by placing the person's foot above the sensing module. A predefined database comparison is made, and the shore output is posted on the internet by using IP address, and also data can be stored with the help of a micro-SD card.

Keywords

Foot Ulcer, Diabetics, Foot Hardness, Temperature Sensor, Arduino Uno

Efficient Audio Noise Reduction System Using Butterworth Chebyshev and Elliptical filter

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Abstract

Digital signal processing is widely used to manipulate, modify, enhance or filter signals such as speech, audio, image and telecommunication signal. Signals can be processed in the analog domain, but the digital domain offers high speed, better accuracy, greater flexibility increased storing capabilities and simple implementation. The audio signal noise reduction has become a fundamental area of research for many real world applications. There are two forms of filters one is fixed, and another one is tunable. Fixed filters measure those within which passband frequencies and stop band frequencies, square measure mounted whereas just in case of tunable filters, passband and stop band frequencies measured variable. These frequencies will be modified consistent with the need of the applications. Tunable digital filters are widely used in medical electronics, digital audio instrumentation, telecommunications and control systems. It is often the essential requirement for removal of noise from the audio signal.

Keywords: Fixed Filters, Tunable Filters, Audio Noise Reduction System

1. Introduction

It is vital for engineers of inserting frameworks, processors, and tools targeting video applications. The audio noise reduction system is the system that's accustomed take away the noise from the audio signal. Audio noise reduction systems are classified into two basic processes. Coming to the primary process is that the complementary sort that involves compressing the audio signal in a well-defined manner before it's recorded. The second method is that the single-ended or non-complementary sort that utilizes techniques to cut back the noise level already placed within the supply material—in essence a playback solely noise reduction system. This method is employed by the LM1894 integrated circuit, is designed for the reduction of hearable noise in nearly any audio source. Noise reduction is the method of removing noise in the signal. All analog or digital recording devices have traits that build them vulnerable to noise [15].

In device's mechanism or processing algorithms having noise. That will be white or random noise without coherence. There can be Active noise control (ANC), also called noise cancellation. Active noise reduction (ANR) may be a technique to reduce unwanted and unprocessed sound by the addition of a second sound specifically designed to cancel the first. The sound may be a pressure wave, or we may say the sound is that the analog signals that are processed consistent with their frequency, that is consisting of a rarefaction phase and compression phase. A noise-cancelling speaker is emitting a sound wave with an equivalent amplitude, however, with inverted phase (also called antiphase) to the original sound signal [9]. The waves are mixed to make the new wave, in an exceedingly process known as interference, and effectively cancel one another out an impression that is named phase cancellation.

Today's advanced active noise control is achieved usually through the employment of analog circuits or digital signal process. An associate adaptive algorithm, the square



International Journal of Electronics >

Volume 104, 2017 - Issue 4

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Original Articles

An efficient pipelined architecture for real-valued Fast Fourier Transform

M. Aravind Kumar ✉ & K. Manjunatha Chari

Pages 692-708 | Received 19 Sep 2015, Accepted 04 Sep 2016, Published online: 17 Nov 2016

🗨 Cite this article 🔗 <https://doi.org/10.1080/00207217.2016.1242165>



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ABSTRACT

Real-valued Fast Fourier Transform (FFT) plays an important role in today's digital world because of the fact that most of the signals contain real values. The FFT computation of real signals using conventional techniques requires more hardware space with high power consumption, which is the most important task for a researcher while designing VLSI architectures. This can be eradicated by clearly analysing the symmetric property of the real-valued signals. In this paper, we have adopted the symmetric property and designed an efficient pipelined architecture for 16-point DIF FFT. The pipeline scheme reduce the processing time at the cost of some registers and in order to contribute efficiently for power reduction we have modified the complex multiplier with reduced internal real multipliers which are in turn replaced by an modified canonic signed digit multiplier (CSDM) with resource-sharing technique. The complete module is synthesised and simulated using Xilinx ISE 14.1 with the target device is Virtex-5 xc5vlx110T. The experimental



Efficient FPGA-based VLSI architecture for detecting R-peaks in electrocardiogram signal by combining Shannon energy with Hilbert transform

Madam Aravind Kumar  Kamsali Manjunatha Chari

First published: 01 August 2018

<https://doi.org/10.1049/iet-spr.2017.0201>

Citations: 18

Abstract

Electrocardiogram (ECG) is a critical application in light of R-peak detection. The R-peaks are impacted by some QRS complex and noises in the current method. Basic testability, snappier execution, and confirmation choices are accomplished by the utilization of the field programmable gate array (FPGA). But, FPGA execution gives less accuracy. To battle execution trouble, another R-peak detector is proposed and relying upon the propelled peak finding logic, which incorporates a Bandpass filter and first-order differentiation process, which are done in the primary phase of the strategy. The noises in the input ECG signal are decreased in the first stage. In the subsequent stage, the smooth Shannon energy envelope (SEE) is acquired by utilizing SEE extraction and the zero phases filtering process. The false R peak is stifled by proposing a Hilbert transform (HT) in the third stage. The HT requires more hardware space and high-power utilization in the current technique. Due to this reason, Real valued Fast Fourier Transform (RFFT) and Inverse RFFT (IRFFT) techniques are proposed in the HT. In the fourth stage, R-peak acknowledgment is evaluated using Massachusetts Institute of Technology, Beth Israel Hospital (MIT-BIH) arrhythmia database and produced the average accuracy of 99.86%, sensitivity of 99.95% and the positive predictivity of 99.90%.

1 Introduction

The field programmable gate array (FPGA) generally utilised as a part of the helpful field application in light of its low-power usage and versatile programming. The FPGA consists of reconfigurable logic, I/O and interconnection blocks, but differs from microcontrollers and Digital Signal Processing (DSP) processors that are Von Neumann type of machines [1].

Madam Aravind Kumar* and Kamsali Manjunatha Chari

Noise Reduction Using Modified Wiener Filter in Digital Hearing Aid for Speech Signal Enhancement

<https://doi.org/10.1515/jisys-2017-0509>

Received October 9, 2017; previously published online April 5, 2019.

Abstract: Speech signals are usually affected by noises during the communication process. For suppressing the noise signal that is combined with the speech signal, a Wiener filter is adapted in digital hearing aids. Wiener filter plays an important role in noise suppression and enhancement by estimating the relation between the power spectra of the noise-affected speech signal and the noise signal. Power consumption and the hardware requirement are the important problems in adapting Wiener filter for major communication systems. In this work, we implemented an efficient Wiener filter and applied it for noise suppression along with a real-valued fast Fourier transform (FFT)/real-valued inverse FFT processor in digital hearing aids. The pipelined process was adopted for increasing the performance of the system. The proposed Wiener filter was designed to remove the iteration problems in the conventional Wiener filter. The division operation was replaced by an efficient inverse and multiplication operation in the proposed design. A modified architecture for matrix inversion with low computation complexity was implemented. The complete design computation was based on IEEE-754 standard single-precision floating-point numbers. The Wiener filter and the whole system architecture was implemented and designed on a Field Programmable Gate Array platform and simulated to validate the results in Xilinx ISE tools. An efficient reduction in power and area was obtained by adapting the proposed method for speech signal noise degradation. The performance of the proposed design was found to be 50.01% more efficient than that of existing designs.

Keywords: Noise, Wiener filter, real-valued FFT, power spectrum, floating point.

1 Introduction

Digital hearing aids are widely used by hearing-impaired people to improve their speech intelligibility and quality of life. However, hearing aid performance is usually degraded due to acoustic feedback, which generates other problems. This phenomenon is produced when the sound propagates from the loudspeaker to the microphone. This causes instability and a high-frequency oscillation that can be perceived by hearing-impaired people if its level exceeds their hearing thresholds. Also, these effects limit the maximum gain that the hearing aid can perform and reduce the sound quality when the gain is close to the limit. To reduce acoustic feedback, several methods based on adaptive algorithms have been used [20] for feedback reduction. A number of techniques are used for speech noise reduction, like non-local diffusion filters [18], acoustic feedback reduction based on finite impulse response and infinite impulse response adaptive filters in digital hearing aids [14], and noise reduction Wiener filter [5]. In the above techniques, the noise is suppressed using filters such as adaptive filters or Wiener filter. Many documented implementations (least mean square, recursive least square, Kalman filters) have been done by using adaptive filters. Adaptive filters are optimal in that they minimize the mean squared estimation error [13] and they can also be computed in real time.

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Bag-of-Surf and Spatial Pyramid Matching for Food Recognition and Calorie Extraction

M. Aravind Kumar, D. Naresh, Srinivas Bachu

Abstract: *Now a days the diet control become habitat to persons all over the world. Hence, in this paper the estimation of calories from recognized food is explained using various algorithms. Basically, the main techniques for extraction of features are bag-of surf method and SPM method and also classification is based on support vector machine classification. The classification rate is about 86% and the results are examined based on the PFID dataset. This technique can also be experimented by using Android.*

Index Terms: SURF, PFID, food recognition, spatial pyramid matching, calorie estimation.

I. INTRODUCTON

Dietary meals vitamins and minerals are the fundamental supply for maintain existence. Food being the essential wellspring of force it's expended in an assortment of style with different introductions [1, 5]. Prior nourishment was once devoured customarily as entire vegetables, natural products, grain, dairy item and nectar. Instruction of any sustenance with the top notch supplement esteems is of vital test both in home and modern condition.

Sustenance classification acknowledgment offers with the mechanized awareness of dinners objects/types with capacities relating to customized dietary appraisal, nourishment recommendation contributions and web-based social networking (suppers) photographs examination. This zone is turning into a thriving report territory with the expansion of reasonable phones and online life administrations. Picture preparing and AI are at the coronary heart of such procedures with different innovations equal to 3D recreation moreover being connected when divide/amount estimation is required to inexact caloric amounts. We suggest two methods to respect food objects: Speeded up robust features (SURF) descriptor established by Bag-of features model and Spatial Pyramid Matching (SPM) [5, 6]. In bag-of-SURF approach, we first build a dictionary of code

words, then generate a histogram of code words for all trained pixels and use linear kernel classification scheme. Spatial pyramid matching procedure tries to account for the spatial information via dividing and subdividing the given food image and developing the histogram of code words of individual areas. We then teach a classifier with spatial pyramid kernel utilizing LIBSVM bundle [7, 8].

II. RELATED WORKS

Herbert Bay et al., [1], proposed novel scale-and pivot invariant scale factor identifier and descriptor, authored as SURF (quickened solid highlights). It approximates and even beats earlier proposed plans with appreciate to repeatability, zone of interests, and power, yet will likewise be processed and in correlation so much quick and adaptable.

That is finished by methods for depending on different pictures for picture convolutions; by methods for developing on the qualities of the dominating surf identifiers and descriptors and with the guide of rearranging these techniques to the prevalent. This outcome in a blend of novel location, portrayal, and coordinating advances.

The paper offers experimental final result on a traditional analysis set, as nice as on imagery got throughout the context of an actual-existence object attention utility. Both methods shows SURF method has strong efficiency.

S Lazebnik, C Schmid, J Ponce [2], presented a technique for perceiving scene classes headquartered on rough world geometric correspondence. This procedure works by methods for using apportioning the picture into increasingly more fine sub-locales and processing histograms of close-by angles found within each sub-provincial. The resulting spatial pyramid is a simple and computationally ground-breaking augmentation of a request less pack-of-surf picture portrayal, and it indicates massively increased effectivity on convoluted scene arrangement obligations. Fundamentally, our proposed methodology surpasses the best in class on the Caltech-one hundred and one database and accomplishes extreme precision on a tremendous database of fifteen normal scene classes.

Chih-Chung Chang and Chih-Jen Lin [3], Proposed LIBSVM library for support Vector Machines (SVMs). We now have been actively opening this bundle in view that the 12 months 2000. The intent is to support persons to effortlessly conform to SVM to their purposes. LIBSVM has got big status in computer imaginative and prescient services. Listed here, we reward all implementation small print of LIBSVM.

Revised Manuscript Received on 30 May 2019.

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Smart City IoT System Network Level Routing Analysis and Blockchain Security Based Implementation

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Received: 11 April 2021 / Revised: 21 July 2022 / Accepted: 16 August 2022 / Published online: 10 October 2022
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Abstract

This paper demonstrates, network-level performance analysis and implementation of smart city Internet of Things (IoT) system with Infrastructure as a Service (IaaS) level cloud computing architecture. The smart city IoT network topology performance is analyzed at the simulation level using the NS3 simulator by extracting most of the performance-deciding parameters. The performance-enhanced smart city topology is practically implemented in IaaS level architecture. The intended smart city IoT system can monitor the principal parameters like video surveillance with a thermal camera (to identify the virus-like COVID-19 infected people), transport, water quality, solar radiation, sound pollution, air quality (O₃, NO₂, CO, Particles), parking zones, iconic places, E-suggestions, PRO information over low power wide area network in 61.88 km × 61.88 km range. Primarily we have addressed the IoT network-level routing and quality of service (QoS) challenges and implementation level security challenges. The simulation level network topology analysis is performed to improve the routing and QoS. Blockchain technology-based decentralization is adopted to enrich the IoT system performance in terms of security.

Keywords IoT technology · Smart applications · Network simulation · Blockchain technology

1 Introduction

Because of its impressive performance and great potential, the IoT technology's role in the design of smart systems is thrived in many fields like smart cities [1–3], medical [4–6], aquaculture [7–9], industry [10–12], and smart home [13]. Industry 4.0 is aimed at mass customization and cyber-physical cognitive systems, in that IoT technology proved its ability and significance. When compared with RFID and smart device technology IoT plays a vital role in exploring smart applications [14].

The IoT device users count is growing rapidly as shown in Fig. 1, and this is the major motivating point for many kinds of researchers to choose IoT as a major domain of research [15]. However, the major research challenges in the implementation of IoT systems are maintaining high throughput, less delay, low power consumption, low path loss, good packet receive rate, quality of service (QoS), congestion

control, reliability, heterogeneity, scalability, high security, and best network routing [16, 17]. The network-level IoT topology simulation helps to make the implementation cost-effective and the incorporation of blockchain technology in IoT will make the system more secure and sturdy [18].

The present-day civil people are using more and more IoT devices and there is a significant demand for smart IoT systems like the design of smart transport, smart building, smart home, smart business [19], and smart grids [20]. However, the development of smart city applications using IoT is one of the potential research areas. Nowadays IoT systems for smart cities monitor viruses (like COVID-19) spread which is very essential. The vital smart city parameters are video surveillance, transport, water quality, solar radiation, sound pollution, air quality (O₃, NO₂, CO, Particles), parking zones, iconic places, E-suggestions, and public relational officer (PRO) information as shown in Fig. 2.

The sensor, network, and implementation level challenges of a smart city-based IoT system as listed in Table 1. Identifying the best IoT routing topology for a smart city with good quality of services (QoS) is one of the network-level potential research challenges. Before the implementation,

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HAND GESTURE RECOGNITION FOR VOCAL IMPAIRED COMMUNITY

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Abstract:

Hand gestures are one of the most prevalent non-verbal and visual communication methods used by people with speech impairments all around the world. The current issue is that most people are unable to understand hand signals or turn them into spoken language quickly enough for the listener to comprehend. The major goal of this project is to provide a better solution for speech impaired persons by creating a glove that will help to eliminate or at least reduce the communication gap between speech impaired and non-speech impaired people. The Arduino Uno serves as the microcontroller in our prototype, which is connected to flex sensors and an accelerometer. for reading hand gesture, LCD display for displaying text output and a Bluetooth module for generating the voice output in mobile.

KEYWORDS: Hand gestures, Flex sensor, accelerometer, Arduino Uno,Text to speech, Bluetooth

1. Introduction:

It was created in response to the necessity for the development of an electronic device that can translate sign language into voice, allowing silent populations to communicate with the broader public. The data gloves are regular textile driving gloves with flex sensors running the length of each finger. Dumb persons can use the gloves to make hand gestures, which are then translated into speech so that normal people can comprehend them. In sign language, a gesture is a distinct movement of the hands that creates a specific shape. The goal of this project is to break down communication barriers. One of the richest sources of tactile sensory data is the hand, which allows for precise and intricate manipulation. Human robot interaction systems that leverage the hand's sensitive manipulation ability have been actively investigated for rehabilitation, virtual reality, entertainment, tele-operation, power support, and other applications. Measurement of unrestricted hand motion should come first in the development of such devices. Hand motion measuring systems have yet to be completely utilised.

Hearing and speech loss can make people feel alone and lonely, which can negatively impact their social and professional lives. They use sign language to express themselves. Sign language is a well-structured code gesture in which each gesture has a specific meaning. For deaf people, sign language is their only form of communication. Many techniques have been developed as science and technology has progressed, not only to reduce the problem of deaf and dumb people, but also to apply it in various industries. Instead of using voice or sound patterns, sign language relies on physical communication and body language to deliver its meaning. This mostly entails a combination of hand forms, orientation, and movement. Sign Language is used not only by the deaf, but also by those who can hear but cannot speak.

We are going to design a system that will interpret sign language and have a playback voice module at the output so that we can Hearing and speech loss can make people feel alone and lonely, which can negatively impact their social and professional lives. They use sign language to express themselves. Sign language is a well-structured code gesture in which each gesture has a specific meaning. For deaf people, sign language is their only form of communication. Many

AUTOMATIC POWER METER READING SYSTEM USING GSM NETWORK

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Abstract:

Electricity is one of the most important requirements for living a comfortable and productive life, and it should be used wisely to ensure its optimal use. Energy Meter Reading system (AEMR) regularly read the energy meter and calculate total amount of bill at the set dead line and sends the message to service provider. From energy meter received data i.e. user name, meter ID, total units with paying amount this message maintained at database server which located at service provider department. With the current energy meter reading and billing system, consumers are dissatisfied with the services provided by power companies. The majority of their complaints are about statistical inaccuracies in their monthly bills. Our project's goal is to build and install an automatic energy meter reading system that measures and transmits total energy consumption to the Electricity Provider via the GSM network, as well as remotely disconnecting and reconnecting the meter. This is accomplished with the help of a microcontroller unit that continuously monitors and records energy meter readings in a permanent memory location, as well as a GSM modem that allows remote monitoring and control of the energy meter. The suggested Microcontroller-based system continuously records readings, and on request, the live meter reading can be relayed to the provider using the GSM network's existing short messaging services (SMS). If necessary, this system can also be utilized to disconnect or reconnect the power supply to the customer. Each energy meter requires a dedicated GSM modem with a SIM card connected to the microcontroller's ports. This saves a lot of energy, and the consumer gets the most out of the electrical energy they pay for; as a result, there is an improvement in the quality, stability, and usage of electrical energy, as well as a reduction in human dependency.

Keywords: Digital Energy meter, GSM module, RTC, Microcontroller, MAX-232

Introduction:

Human operators perform a substantial amount of the meter reading for power use, moving from building to building to collect data for bill generation. The human operator must also visit the premises to disconnect the electricity supply in the case of consumers who do not pay their bills. To cover the requisite area of interest, a large number of operators and lengthy working hours are required. This method of meter reading for invoicing is prone to human mistake and takes a long time. Weather conditions may limit or stifle the operation. This billing system is inaccurate, inefficient, and raises the cost of meter reading for the energy provider. . The prepaid electronic energy meter was recently introduced, largely resolving the issue of having to visit the site to take meter readings. However, several issues remain, such as the prepaid card's vulnerability to damage or loss, the inability to provide real-time monitoring, and the inability to prevent theft. With the rapid improvement in wireless communication technology and microelectronics, the number of power consumers and commercial operations in the electrical industry has increased significantly. In order to accomplish efficient meter reading, eliminate billing mistake, and lower operational costs, it became increasingly vital to develop automatic electric energy meters to replace existing manual energy meters. Automatic meter reading is a cost-effective data gathering method that offers increased data accuracy, allows for more frequent readings, enhanced invoicing and customer service, timely energy

IoT BASED GREENHOUSE MONITORING AND CONTROLLING SYSTEM

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Abstract

The goal is to create a smart greenhouse with a regulated environment in which to grow plants. Utilizing a low-cost, high-efficiency programmable module to detect climatic behaviour within the greenhouse and adjust settings based on crop production needs, using a variety of methodologies and the board ESP8266 Node MCU module. The water content of the soil, the light intensity originating from natural or artificial sources, and the temperature and humidity of the field area are the parameters that need to be optimised. all of these sensors capture data and send it to the NodeMCU module, which processes it and controls all of the parameters via the water pump, motors, exhaust system, and lighting system based on the data computations. The HTTP protocol is used to link the NodeMCU module to a wireless internet connection or IoT platforms such as telegram bot. The acquired environmental parameters data was supplied to farmers' smartphones via internet mode, allowing them to keep a proper eye on their farms no matter how far away they were. In the sphere of agricultural and food production, technology has advanced at a breakneck pace and continues to do so. in the field of agriculture, to optimise and obtain maximum plant growth In the area of Android/IDS smartphone applications, an accurate system would undoubtedly bring about change.

Keywords: NODE MCU,Relay,GSM module, temperature and humidity sensor (LM35), Moisture sensors, smart phones, DC motors.

1. Introduction

A greenhouse is a closed structure that protects plants from environmental variables such as temperature conditions, pollution, and so on. It allows plants to grow in a sustainable and effective manner throughout the year.

Sunlight, water content in the soil, temperature, humidity, and other basic elements all influence plant growth. The automated water delivery system for urban residential areas, according to an article, can be utilised to effectively manage water resources. Because the required physical parameters are difficult to control manually inside a greenhouse, an automated system is required. Many smart irrigation systems have been proposed and constructed using technologies such as evapotranspiration (ET), thermal imaging, capacitive methods, neutron scattering technology, and gypsum blocks, to name a few. Capacitive \ssensors Instantaneous, on the other hand, are expensive and must be calibrated frequently with changes in temperature and soil type [Our suggested system in this paper collects three parameters from the sensors and activates the actuators if the real values are greater than the threshold values, as well as storing these values in a cloud database that can be accessed from anywhere, at any time, require chilly temperatures to grow. Cucumbers, melons, and other summer crops require moderate to hot climatic conditions to thrive. Moisture sensors, temperature and humidity sensors, an Arduino Uno, and water pipes to supply water from a tank controlled by DC motors make up the prototype we used.

Moisture sensors (YL 69) are placed near the roots, while a temperature and humidity sensor (LM35) is placed further away to detect temperature and humidity and evaluate the data. If the temperature and humidity levels are higher than the reference value, collingfan will turn on to keep them within the threshold levels.

**INDUSTRY PARAMETERS MONITORING AND CONTROLLING SYSTEM BASED
ON EMBEDDED WEB SERVER**

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Abstract

The Internet of Things (IoT) is a technology that is quickly advancing. The Internet of Things (IoT) is a network of physical objects or things that are equipped with electronics, software, sensors, and network connectivity, allowing them to collect and exchange data. Using wireless devices, Android, and sensors, we are developing a system that will automatically analyse industrial applications and produce a means to build powerful industrial systems and apps. Sensor data acquisition for industrial Wireless Sensor Networks (WSN) in an Internet of Things (IoT) environment requires a sensor interface device. The goal is to provide a reconfigurable sensible device interface for industrial WSNs in an IoT environment, with the ESP 8266 12E as the core controller. As a result, it will scan information on numerous completely distinct devices in simultaneously and in real time at a fast pace. This style uses the Intelligent Device Interface specification. The device features the most up-to-date ARM programmable technology as well as an intelligent device standard. It can quickly determine the temperature by detecting the values of sensors. Voltage, current status of real-time measurement, and humidity and gas presence in the industrial area It also has irregularities that it uses to manage the power. In order to avoid a critical crisis and properly implement preventive actions. It is the most efficient and cost-effective method of monitoring equipment safety. As a result, it has excellent social prospects.

KEYWORDS: Wireless Sensor Networks (WSN), Internet of Things (IoT), Sensors, Sensor data acquisition, ESP 8266 12E Control

1. Introduction:

To meet industrial objectives ,industry Standards must be adhered to, which are concerned with the proper functioning and execution of operations in their respective domains [1].

Global and domestic competitiveness are aided by industry standards. It's a critical instrument for setting and achieving industry objectives. Today's producers must accomplish more with less in order to meet the stated objectives. Reduced overheads and improved quality throughput. Proactively detecting and reacting to problems before they worsen is critical. Learn from the process and use what you've learned to your next project. These are the key parameters of a successful industry, and they may be attainable if machine and manufacturer communication is accurate. With the support of a real-time monitoring system, this type of accurate communication is achievable. Users can use this system to continuously monitor the process and gather real-time information about the equipment's reliability, accessibility, and maintainability.

Furthermore, not only is this data collected, but it is also shared swiftly and precisely in order to meet the high production objective. In comparison to Supervisory Control and Data Acquisition, this monitoring facility provides a solution that is not only powerful but also cost-effective.

is an essential consideration for the industry. In this study, a real-time data monitoring system is investigated



Radix 2/24 and 2/25 Split Radix Fast Fourier Transform Designs for Reduced Multipliers with Twiddle Factor Approximation

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Abstract:

This work proposes a novel method for designing split-radix Fast Fourier Transform architecture. The new Radix- 2/24 and Radix- 2/25 data flow algorithms that are proposed in this paper have effective twiddle factor multiplier reduction. FFT with the radix of split initially modified so that all the twiddle multipliers are placed at two stages and the less important multiplier of multiplication with w_81 replaced by two-point butterfly followed by shift register. This technique significantly reduces the twiddle factor of w_81 multiplication, which in turn lowers the overall multipliers for higher order FFT. Three real multipliers are used to produce complex multiplication, and five real additions are used to further reduce multipliers. Comparing the proposed data flow designs to other recent existing ones will result in lower overall multipliers. The proposed designs use less power than other recent existing designs.

Keywords—Adders; Split-Radix; FFT; Multipliers; Shift Register; Twiddle factor

1.INTRODUCTION

Among the many transformations used in signal processing, the Discrete Fourier transform (DFT) has a great deal of application in the fields of communication, image processing, and signal processing. In all these fields, DFT cannot be computed directly from a mathematical formulation due to computations in the real world. FFT is one who compute DFT in a very efficient manner, in the perspective of adders and multipliers, FFT algorithms with different variations in radix [1], [2], [3] is much significant research area, this algorithms grows rapidly after invention of Cooley and turkey [4] who are behind the invention of the FFT. The discrete sequence x_n has the DFT is X_k expressed in the equation (1).

$$X_k = \sum_{n=0}^{N-1} x_n W_N^{kn} \quad (1)$$

Radix-2 algorithm is the basic for any higher radix algorithm, in DIF FFT X_{2k} and X_{2k+1} is expressed in equations (2) and (3), the flow of radix-2² DIF FFT is shown in Figure 1 in which twiddle factor multiplication is done for every two stages.

FAST IMAGE ENCRYPTION BASED ON RANDOM IMAGE KEY

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Abstract

Internet plays an important role in circulating a huge amount of multimedia. An example of this multimedia is the image. To send an image over the network secretly, the sender tries to find encryption algorithm to hide image information. This paper aims at designing an efficient encryption algorithm for color image using random image key generated with minimum time execution for encryption and decryption operations. XOR operation is used here to make more diffusion of the encrypted image to maintain a higher level of security upon transference than it is with the original image.

Keywords – Encryption, Decryption, Random Key, XOR operation.

1. INTRODUCTION

Network technologies and media services provide ubiquitous conveniences for individuals and organizations to collect, share, or distribute images/videos in multimedia networks and wireless or mobile public channels. Image security is a major challenge in storage and transmission applications. As a matter of fact, all users, who use multimedia such as image, audio, video and text, may need to protect information from attacks during sending or receiving them through channel. There are two challenges for multimedia encryption; the first one is the size of data and the second is the cost of encryptions. In this paper, an image encryption method based on a new random key generated from the same image is going to be adopted.. Image Crypto system can be classified into two main sections; one for encryption and the other for decryption. In this paper a new algorithm is proposed to encrypt color image using symmetric key which is generated from the same image or any image can be selected. Some tests are applied here to determine performance algorithm. These are histogram, mean square error, peak signal to noise ratio, entropy, correlation coefficients, number of changing pixel rate and unified averaged changed intensity. The proposed algorithm was satisfied with good results where speed of running was good for encryption and decryption algorithm.

2. RELATED WORK

In this section many studies are summarized here to survey some ideas about the image encryption during the last years. Pratibha S.Ghode et al improved a keyless method for image cipher in lossless color images to encrypt and decrypt image without any loss of data quality. Khanzadi H. et al proposed an image encryption algorithm using bit sequence random generator based on Chaotic Logistic and Tent maps. Mirzaei et al introduced a new parallel algorithm for image encryption. First of all, the plain image is divided into 4 equal blocks and then the position of each block is shuffled. Then a total shuffling algorithm is applied to the whole image. After this, we use different values for encrypting each pixel in each of the 4 blocks of the whole image.

SMART CONTROL OF TRAFFIC LIGHT USING DEEP LEARNING

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Abstract

Traffic congestion is becoming one of the critical issues with increasing population and automobiles in cities. Traffic jams not only cause extra delay and stress for the drivers, but also increase fuel consumption and air pollution. Although it seems to pervade everywhere, megacities are the ones most affected by it. And its ever-increasing nature makes it necessary to calculate the road traffic density in real-time for better signal control and effective traffic management. The traffic controller is one of the critical factors affecting traffic flow. Therefore, the need for optimizing traffic control to better accommodate this increasing demand arises. This paper is utilizing traffic cameras and YOLO object detection algorithms to estimate traffic density at all lanes and then adjust red and green signal time. Cameras will take snapshot of all lanes every five seconds and then estimate traffic at lanes and based on density green and red signal time will be adjusted.

Keywords – Traffic control, Traffic light system, Traffic management, Intelligent transport systems, Smart surveillance, Computer Vision, Machine Learning, Object detection, YOLO.

1. INTRODUCTION

In recent years, video monitoring and surveillance systems have been extensively used in traffic management for security, ramp metering, and providing information and updates to travellers in real-time. The traffic density estimation and vehicle classification can also be achieved using video monitoring systems, which can then be used to control the timers of the traffic signals so as to optimize traffic flow and minimize congestion. Our proposed system aims to design a traffic light controller based on Computer Vision that can adapt to the current traffic situation. It uses live images from the CCTV cameras at traffic junctions for real-time traffic density calculation by detecting the number of vehicles at the signal and setting the green signal time accordingly.

2. RELATED WORK

An intelligent video-based drowsy driver detection system, which is unaffected by various illuminations, is developed in this study. Even if a driver wears glasses, the proposed system detects the drowsy conditions effectively. By a near-infrared-ray (NIR) camera, the proposed system is divided into two cascaded computational procedures: the driver eyes detection and the drowsy driver detection. The average open/closed eyes detection rates without/with glasses are 94% and 78%, respectively, and the accuracy of the drowsy status detection is up to 91%. By implementing on the FPGA-based embedded platform, the processing speed with the 640×480 format video is up to 16 frames per second (fps) after software optimizations.

A vision-based real-time driver fatigue detection system is proposed for driving safely. The driver's face is located, from colour images captured in a car, by using the characteristic of skin colours. Then, edge detection is used to locate the regions of eyes. In addition to being used as the

FORECASTING ENERGY CONSUMPTION USING ARIMA MODELS

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Abstract

The paper is aimed at contributing to the body of knowledge that exist in the area of energy forecasting by reviewing relevant empirical works on energy forecasting using ARIMA models. This paper is relevant in the face of frequent power outage and the dependence on external economies for energy supply. The study is based on secondary data obtained from electronic journals through archival studies. In all 10 articles were selected through purposive sampling method and were analysis using content analysis method. The results indicate that future energy consumption is expected to increase in economies in which these forecasts have been done. Hence, energy use must be efficient to avoid energy crisis in future. Future research should look at review of works on forecasting in a comparative manner comparing other models that have been used in forecasting energy demand. The paper is limited by the use of only secondary data.

Keywords: Forecasting, ARIMA model, Energy consumption, Disaggregate energy consumption

1. INTRODUCTION

Forecasting of energy consumption for both domestic and non-domestic purposes is of national interest for short run and long run economic policies.

The intractable energy crisis in many economies including Ghana has made the forecasting of energy consumption relevant and as such the need to get appropriate forecasting model is very crucial (Mohamed and Bodger[2], 2005). The consumption of energy has been increasing over the years due to factors such as population growth and technology development which aim at improving the wellbeing of citizens. Ghana like other developing economies over the years has been faced with energy crisis at the domestics and industrial levels with cases of total blackout in the country.

2. RELATED WORK

Many models have been used to forecast energy consumption empirically. In some economies energy consumptions has been increasing while in other economies energy consumption has been going down. According to Ajith and Baikunth (2001)[1] if energy consumption is more than production energy crisis may occur. This means to prevent future energy crisis production should not lag behind energy consumption.

Wang and Meng (2012) [3] predicted energy consumption of Hebei province from 2009 to 2013. The results demonstrate that the energy consumption in Hebei province will continue to increase for the next 5 years.

FACIAL EMOTIONS RECOGNITION USING CNN

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Abstract

Nowadays, deep learning techniques know a big success in various fields including computer vision. Indeed, a convolutional neural networks (CNN) model can be trained to analyze images and identify face emotion. In this paper, we create a system that recognizes students' emotions from their faces. Our system consists of three phases: face detection using Haar Cascades, normalization and emotion recognition using CNN on FER 2013 database with seven type of expressions. Obtained results show that face emotion recognition is feasible consequently, it can help to modify their presentation according to the emotions.

Keywords:—facial expression, Emotion recognition, Convolutional neural networks (CNN), Deep learning, Intelligent system

1. INTRODUCTION

Facial expression recognition identifies emotion from face image, it is a manifestation of the activity and personality of a human. In the 20th century, the American psychologists Ekman and Friesen [2] defined six basic emotions (anger, fear, disgust, sadness, surprise and happiness), which are the same across cultures. Facial expression recognition has brought much attention in the past years due to its impact in clinical practice, social robotics and education. The purpose of this article is to implement emotion recognition in education by realizing an automatic system that analyzes students' facial expressions based on Convolutional Neural Network (CNN), which is a deep learning algorithm that are widely used in image classification. It consists of a multi-stage image processing to extract feature representations. Our system includes three phases: face detection, normalization and emotion recognition that should be one of these seven emotions: neutral, anger, fear, sadness, happiness, surprise and disgust.

2. RELATED WORK

Many researchers are interested in improving the learning environment with Face Emotion Recognition (FER). Tang et al. proposed a system which is able to analyze facial expressions in order to evaluate classroom teaching effect. The system is composed of five phases: data acquisition, face detection, face recognition, facial expression recognition and post-processing. The approach uses K-nearest neighbor (KNN) for classification and Uniform Local Gabor Binary Pattern Histogram Sequence (ULGBPHS) for pattern analysis. Savva et al. proposed a web application that performs an analysis of students' emotion who participating in active face-to-face

MISSING CHILD IDENTIFICATION SYSTEM USING DEEP LEARNING AND MULTICLASS SVM

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Abstract

Every year, uncountable numbers of missing children in India are reported. A significant portion of missing children cases have unidentified youngsters. With the use of facial recognition, this study describes a novel application of deep learning approach for locating the reported missing child among the many children's images that are now available. A shared platform allows users to post pictures of questionable children along with notes and landmarks. The image will be automatically compared to the repository's registered images of the missing child. The input child image is categorised, and the missing children database will be searched for the photo that matches the input child image the best. This is accomplished by training a deep learning model to recognise the missing child from the database of missing children. In order to do this, a deep learning model is trained to accurately identify the missing child using the facial image uploaded by the public and the missing child image database that is provided. Here, face identification is accomplished using the Convolutional Neural Network (CNN), a very successful deep learning technology for image-based applications. The classification performance achieved for child identification system is 99.41%. It was evaluated on 43 Child cases. A deep learning [1] architecture considering all these constrain is designed here. The proposed system is comparatively an easy, inexpensive and reliable method compared to other biometrics like finger print and iris recognition systems.

Keywords— Missing child identification, face recognition, deep learning, CNN, VGG-Face, Multi class SVM.

1. INTRODUCTION

The greatest resource in every country is its children. The proper raising of a nation's children determines its future. Children make up a substantial portion of the population in India, the second most populous nation in the world. But sadly, many children go missing every year in India for a variety of causes, such as kidnapping or abduction, runaway children, trafficking children, and misplaced children.

While 174 children go missing in India on average every day, half of them are still unaccounted for, which is a highly unsettling statistic. Children who disappear could be utilised for numerous forms of abuse and exploitation. In accordance with the National Crime Records Bureau (NCRB) study, which the Ministry of Home Affairs (MHA) stated in the Parliament (LS Q no. 3928, 20-03-2018), more than one lakh children (1,11,569 in actual numbers) were reported to have gone missing till 2016, and 55,625 of them remained untraced till the end of the year. Many NGOs claim that estimates of missing children are much higher than reported. Police departments get many reports of missing children. For a variety of circumstances, a child reported missing from one area may turn

FAKE CURRENCY DETECTION USING IMAGE PROCESSING AND DEEP LEARNING

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Abstract

The one important asset of our country is Bank currency and to create discrepancies of money miscreants introduce the fake notes which resembles to original note in the financial market. In general, by a human being, it is very difficult to identify forged note from the genuine not instead of various parameters designed for identification as many features of forged note are similar to original one. To discriminate between fake bank currency and original note is a challenging task. So, there must be an automated system that will be available in banks or in ATM machines. To design such an automated system there is need to design an efficient algorithm which is able to predict whether the banknote is genuine or forged bank currency as fake notes are designed with high precision.

Keywords –Support Vector Machine, Bank currency, Supervised Machine Learning

1. INTRODUCTION

Financial activities are carrying out in every second by many persons in which one most important asset of our country is Banknotes [3]. Fake notes are introduced in the market to create discrepancies in the financial market, even they resemble to the original note. Basically they are illegally created to complete various task [12]. In 1990 forgery issue is not much of concern but in late 19th century forgery has been increasing drastically [13]. In 20th century technology is increasing very vastly that will help the fraudsters to generate fake notes whose resemblance is like genuine note and it is very difficult to discriminate them [1]. This will lead to financial market to its lowest level. To stop this and to conduct smooth transaction circulation forged bank currency must be conserved [16]. As a human being it is very difficult to identify between genuine and forged bank currency. Government have designed banknote with some features by which we can identify genuine [9]. But frauds are creating fake note with almost same features with nice accuracy that make it very difficult to identify genuine note [5]. So, now a days it is required that bank or ATM machines must have some system that can identify the forged note from the genuine note [12].

To determine the legitimacy of the banknote artificial intelligence and Machine learning (ML) can play a vital role to design such a system that can identify forged note from the genuine bank currency [6,7,12].

Now a days, supervised machine learning (SML) approaches for classification problem is widely used. For medical disease it shows even promising results [2]. Few authors have only applied SML algorithms on bank currency authentication [6-9, 12]. To identify whether a note is genuine or fake we have to develop an automation system. Initially, the input is an image of note and from different image processing techniques we can extract the features of note. Further these images are given as an



IMAGE STEGANOGRAPHY USING LSB ALGORITHM

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Abstract:

Steganography is defined as the study of invisible communication .It usually deals with the way of hiding the information of the existence of the communicating data .It hides the facts of information. It is the process of hiding the data from one digital media to another digital media and recover the same information afterwards .This paper focuses on a process of hiding data to image by using the least significant bit(LSB) and the symmetric key between the sender and the receiver. Here we have to choose the bits that will get the minimum resolution between the original image and stego image. This paper further explains how the encryption and decryption processes are done..

Index Terms: LSB, PSNR, Steganography and Dynamic Symmetric Key.

1. INTRODUCTION

Steganography is derived from the two Greek words Stego and Graphia, Stego means covering and graphia which means writing , thus the translation is covered writing or the hiding the data. The simplest way to do this process is by inserting the confidential data bits in LSB positions of original image.[1].

1.1 Types of Steganography:

- Image to image
- Text to image
- Image to text
- Video to voice
- Voice to video

Image to Image :-

In the image steganography the image is inserted within the another image by using the stego key[2].

Text to Image :-

In this, the text is inserted within the image and sends the image with the help of the symmetric key.



IOT BASED VIRTUAL DOCTOR ROBOT

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Abstract

Healthcare monitoring real time data transferring system in hospitals and other health sectors has tremendous growth, and these healthcare monitoring system having advanced technologies becoming one of the best approach across globe nowadays. The system using smart technology in iot sector for monitoring the patient's basic health parameters which are required for the doctors for further treatment. In this system sensors are used to capture the real time data of basic parameters such as temperature, O₂ level and heart rate of the patient from hospital environment. In older days the Traditional method in order to measure all the basic parameters the doctor needs to visit the patient's ward. So there are major problems that the doctor's have to face while checking the patient is that the doctor needs to be present physically at the patients location and second major problem is that if the patient is having contagious disease so it is harmful for the doctor's health also. So in order to prevent these problems we need to have such a system that can collect all the required parameter data i.e. real time data without any contact with the patient we need IoT based virtual doctor robot it is nothing but the parameter monitoring system that can monitor basic parameters such as human body temperature, heart rate and oxygen level of the patient for further suitable treatment. It includes connecting the patient with the doctor, collecting and transferring real time data and monitoring the collected data. To alert the doctor with patient's current status and information. Adopting this strategy in these regions not only contributes to health emergencies, epidemics and pandemics like COVID. But also increases survival rates. Our whole system is controlled and monitored by microcontrollers and also by Internet of Things

Keywords: Virtual Doctor Robot, dispense medicines prescription, Health parameters.

1. INTRODUCTION

Technology in biomedical field is the blessing for human society. But we can see in our daily routine most of the people don't take care of their health seriously. Problems such as cancer and various diseases due to which death rate is increasing because of poor medical infrastructure. So for better observation of health at personal level medical kits are best option. In older days due to lack of technology and facilities doctors were not able to attend all the patient on time, so there was chances of patient death. But nowadays because of this technology doctors can connect with the patient virtually. The doctors can collect all the basic parameters which are required for their treatment so if there is an emergency the doctors can take preventive action instantly so chances of deaths are reducing. As we can see technology like iot is completely dominating the future for development of health sector, iot is nothing but a blessing for us which connects humans with the technology. It includes uses of various

LOW COST IOT BASED SMART AGRICULTURE SYSTEM

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Abstract

In this project, we are going to build a Smart Farming System using IoT. The objective of this project is to offer assistance to farmers in getting Live Data (Temperature, Humidity, Soil Moisture, Soil Temperature) for efficient environment monitoring which will enable them to increase their overall yield and quality of isture sensor, Water Pump, and 12V led strip. When the IoT-based agriculture monitoring system starts, it checks the Soil moisture, temperature, humidity, and soil temperature. It then sends this data to the IoT cloud for live monitoring. If the soil moisture goes below a certain level, it automatically starts the water pump. We previously build Automatic Plant Irrigation System which sends alerts on mobile but doesn't monitor other parameters. Apart from this, Rain alarm and soil moisture detector circuit can also be helpful in building Smart Agriculture Monitoring System.

Keywords : Internet of things, Soil nutrients, Colorimetric principle Sensor network, Fuzzy system.

1. INTRODUCTION

Agriculture is the unquestionably the largest livelihood provider in India. With rising population, there is a need for increased agricultural production. Over the past 15 years, farmers started using computers and software systems to organize their financial data and keep track of their transactions with third parties and also monitor their crops more effectively [1]. In the Internet era, where information plays a key role in people's lives, agriculture is rapidly becoming a very data intensive industry where farmers need to collect and evaluate a huge amount of information from a diverse number of devices (eg., sensors, faming machinery etc.) in order to become more efficient in production and communicating appropriate information [2].

2. RELATED WORK

The term "Internet of Things" refers to the connection of objects, equipment, vehicles, and other electronic devices to a network for the purpose of data exchange (IoT). The Internet of Things (IoT) is increasingly being utilised to connect objects and collect data. As a result, the Internet of Things' use in agriculture is crucial. The idea behind the project is to create a smart agriculture system that is connected to the internet of things. This system's microcontroller is a Raspberry Pi. The temperature and humidity in the surrounding region, as well as the moisture level of the soil, are monitored using the NPK and soil moisture sensor. The data will be available on both a smartphone and a computer. As a result, Internet of Things (IoT) and Raspberry Pi-based Smart Agriculture Systems have a significant impact on how farmers work. It will have a good impact on agricultural productivity as well.



BRAIN TUMOUR IMAGE SEGMENTATION USING DEEP NETWORKS

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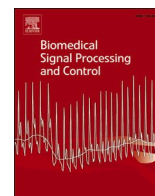
Abstract

Automated segmentation of brain tumour from multimodal MR images is pivotal for the analysis and monitoring of disease progression. As gliomas are malignant and heterogeneous, efficient and accurate segmentation techniques are used for the successful delineation of tumours into intra-tumoural classes. Deep learning algorithms outperform tasks of semantic segmentation as opposed to the more conventional, context-based computer vision approaches. Extensively used for biomedical image segmentation, Convolutional Neural Networks have significantly improved the state-of-the-art accuracy on the task of brain tumour segmentation. In this paper, we propose an ensemble of two segmentation networks: CNN and a U-Net, in a significant yet straightforward combinative technique that results in better and accurate predictions. Both models were trained separately on the BraTS-19 challenge dataset and evaluated to yield segmentation maps which considerably differed from each other in terms of segmented tumour sub-regions and were ensemble variably to achieve the final prediction. The suggested ensemble achieved dice scores of 0.750, 0.906 and 0.846 for enhancing tumour, whole tumour, and tumour core, respectively, on the validation set, performing favourable in comparison to the state-of-the-art architectures currently available.

Keywords—Deep learning, BraTS, Medical segmentation, U-Net, CNN, ensembling.

1. INTRODUCTION

Accurate segmentation of tumours through medical images is of particular importance as it provides information essential for analysis and diagnosis of cancer as well as for mapping out treatment options and monitoring the progression of the disease. Brain tumours are one of the fatal cancers worldwide and are categorised, depending upon their origin, into primary and secondary tumour types. The most common histological form of primary brain cancer is the glioma, which originates from the neural cells and attributes towards 80% of all malignant brain tumours. Gliomas can be of the slow-progressing low-grade (LGG) subtype with a better patient prognosis or are the more aggressive and infiltrative high-grade glioma (HGG) or glioblastoma, which require immediate treatment. These tumours are associated with substantial morbidity, where the median survival for a patient with glioblastoma is only about 14 months with a 5-year survival rate near zero despite maximal surgical and medical therapy. A timely diagnosis, therefore, becomes imperative for effective treatment of the patients. Magnetic Resonance Imaging (MRI) is a preferred technique widely employed by radiologists for the evaluation and assessment of brain tumours. It provides several complementary 3D MRI modalities acquired based on the degree of excitation and repetition times, i.e. T1-weighted, post-contrast T1-weighted (T1ce), T2-



BMO based improved Lite Swin transformer for brain tumor detection using MRI images

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ARTICLE INFO

Keywords:

Convolution neural network
Brain tumour
Lite Swin transformer
eBarnacle's mating optimizer
Fully connected Network
Magnetic resonance images

ABSTRACT

Magnetic resonance imaging (MRI) of the brain is the gold standard for diagnosing degenerative neurological conditions such as brain tumours, strokes, dementia, and Multiple sclerosis (MS). For the purpose of health monitoring, MRI scans, a plethora of deep learning-based medical image processing approaches have been presented. After a fully linked network, a convolution neural network (CNN) is the next best structure. While it excels at improving local features and is translation invariant, it has trouble getting global features. In order to make up for CNNs' flaws and efficiently get global characteristics, transformers can be used. Unfortunately, the calculated number of transformers is too high. An improved Lite Swin transformer (OLiST) is offered as a viable solution to this issue. msOur proposed OLiST model takes advantage of both the global feature acquisition capability of the Lite Swin transformer and the capability of the CNN by combining the features extracted by the two and then sampling them up gradually. The barnacles mating optimizer (BMO) for hyper-parameter optimization in the LiST model enhances its performance in detecting brain tumours. Kaggle database photos of open-source brain tumours re used. Testing the suggested deep learning perfectly with other ll-known transfer learning approaches on the same MRI dataset resulted in an increase in processing time and improved classification results.

1. Introduction

The human CNS involves the cord [1]. The brain is the most significant share of the humanoid nervous scheme. The brain is responsible for evaluating, integrating, organizing, determining, and issuing directives to the break of the body, among other things. The human brain is anatomically extraordinarily complex [2]. Stroke, infection, brain tumours, and headaches are all examples of CNC illnesses that are notoriously difficult to diagnose, evaluate, and find an effective therapy for. When aberrant cells gather in the rigid skull that encases the brain, the result is a brain tumour [3]. Any kind of growth in such a confined space is likely to cause difficulties. Brain damage is certain when a tumour forms inside the skull, making brain tumours a particularly dangerous condition. Brain tumours are the tenth leading cause of mortality in both adults and children [4]. Tumours vary widely in texture, location, and form, yet all have dismal 5-year survival rates.

About 250,000 people each year are diagnosed with a brain tumour, and only 2 % of those are malignant [5]. In 2020, 23,890 individuals in the United States are predictable with a brain tumour; this includes 13,590 males and 10,300 women. There re projected to be 1,879 new cases of brain cancer in Australia in 2020. About 70 % of all cases of primary brain tumours in the United States are found in children under the age of 18. Primary brain tumours have lasting consequences [6], but there is currently no effective treatment for detecting them early. The worldwide incidence of brain tumours grew from around 10 % in 2004 to around 15 % in 2020.

Tumours of the brain and central nervous system (CNS) come in a wide variability of types and stages, from tremendously rare to common [7]. Chief and secondary tumours account for half of the 130 cases of brain cancer.

Prime brain tumours are those that start in the brain itself. A chief brain tumour originates in brain tissue and may spread to the encasing

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<https://doi.org/10.1016/j.bspc.2024.106091>

Received 11 August 2023; Received in revised form 10 January 2024; Accepted 3 February 2024

Available online 13 February 2024

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Biomedical signal compression using deep learning based multi-task compressed sensing

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Article Info

Article history:

Received Jul 4, 2023

Revised Sep 26, 2023

Accepted Oct 19, 2023

Keywords:

Generalization

MIMIC-II

Multi-task compressed sensing

Reconstruction

Robustness

ABSTRACT

Real-time transmission of biomedical signals is immensely challenging and requires cloud and internet of things (IoT) infrastructure. Security is also an important factor; however, to accomplish this, a reconstruction method is developed in which the entire signal is supplied as an input, the primary portion is considered here, and the signal is further encoded and transmitted to the destination. Electrocardiogram (ECG) compression for the lightweight wireless network is quite challenging for long-term healthcare monitoring. Compressed sensing (CS) involves efficient encoding mechanisms for error rate estimation for reconstruction and energy consumption for wireless transmission of data. We propose a multi-task compressed sensing (MT-CS) reconstruction mechanism in this study for ECG compression of data is most chosen for a wireless network system that has various sensors embedded in it. This model further extracts the essential adaptive features for correlation existing in the ECG signals. The performance of the proposed MT-CS reconstruction mechanism is evaluated on the multiparameter intelligent monitoring in intensive care (MIMIC-II) dataset, which ensures its robustness and generalization. The results obtained upon simulation ensure that the proposed MT-CS based reconstruction approach ensures excellent reconstruction signal with fewer measurements in comparison with the existing state-of-art CS techniques.

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1. INTRODUCTION

There is a rapid increase in the internet of things (IoT) being utilized in various fields, however, in the certain domain that includes healthcare, the use of IoT seems to be at a slower pace [1]. IoT in the medical domain has a collection of people as well as equipment that require wireless communication systems for the interchange of data on healthcare, and monitoring of patients resulting in a better-improved quality of patient healthcare. Healthcare IoT improvises the quality of patient life as well as creates an improved cost-efficient environment and higher quality of services. The signal electrocardiogram (ECG) is an essential signal that is recorded by the use of medical equipment, it detects and amplifies signals occurred at heartbeats. Various heart abnormalities can be detected by the use of these signals that could be fatal [2], [3]. The sampling of these signals is done for frequencies more than 100 Hz. Whereas the abnormality in beats seldom varies, hence these signals have to be recorded and studied for an extended period. Recently, an algorithm that is energy-saving and rapid is developed for compressive sensing (CS) that is utilized for

IoT driven joint compressed sensing and shallow learning approach for ECG signal-reconstruction

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Article Info

Article history:

Received Sep 11, 2023

Revised Oct 18, 2023

Accepted Jan 11, 2024

Keywords:

Generalization

MIMIC-II

Multi-task compressed sensing

Reconstruction

Robustness

ABSTRACT

Because biological signal transmission in real time might be very demanding, cloud and internet of things (IoT) infrastructure are required. To do this, the main component of the signal serves as the focal point of a reconstruction strategy that has been developed. The input is transferred to the intended destination once it has been encoded. Security is an important consideration that must not be disregarded. For long-term healthcare monitoring via lightweight wireless networks, electrocardiogram (ECG) compression is a major difficulty. Reducing energy consumption in wireless data transmission and precisely calculating error rates for data reconstruction are two essential components of compressed sensing. The application of effective encoding methods is crucial for these considerations. We present multi-task compressed sensing (MT-CS), a unique method for compressing ECG data. When used to wireless network systems with several embedded sensors, this technique is quite effective. From the ECG data, the model learns the fundamental adaptive properties needed for correlation. We use the multiparameter intelligent monitoring in intensive care (MIMIC-II) dataset to investigate the performance of the suggested MT-CS reconstruction technique in order to assess its strength and application. In comparison to current compressed sensing methods, the simulation results demonstrate that the suggested reconstruction methodology utilizing MT-CS generates high-quality reconstruction signals with fewer observations.

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1. INTRODUCTION

The internet of things (IoT) is becoming more and more common in many businesses, but its adoption in the healthcare industry seems to be happening more slowly [1]. IoT is utilized in the medical profession to build wireless communication networks that make data sharing for patient monitoring and treatment easier. This technology is essential to raising the standard of patient care. IoT integration in healthcare has shown to have major benefits, improving patient outcomes and creating more effective, cost-effective environments that offer top-notch services. One important signal that is recorded using medical equipment is the electrocardiogram, or electrocardiogram (ECG) signal. Heartbeat pulses are detected and amplified by this technique. These signals are capable of identifying a variety of potentially fatal cardiac anomalies. Beat consistency is something that they seldom stray from. Thoroughly capturing and analyzing



A Chest X-Ray Image Based Model for Classification and Detection of Diseases

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Abstract. Radiography holds significant importance in the medical field, allowing for efficient and widespread access, primarily due to the predominant utilization of thoracic imaging equipments within healthcare infrastructure. However, radiologist's ability to understand radiography images is constrained by their inability to recognize the fine visual details present in the images. There have been numerous studies published in the literature describe machine learning (ML) recent advance models that use support vector machines do differentiate between COVID-19 and non COVID-19 cases by means of open-access chest radiograph databases. The AI can be familiar with characteristics observed in chest X-rays tasks that are typically beyond the scope of a radiologist. They did, however, produce poor categorization performance. Deep - learning techniques in artificial intelligence (AI) are high-performance classifiers engage in recreation a most important significance in the identification of the disease through analysis of chest radiograph metaphors. The major goal of this learning is to examine the enhancement of pre-trained ConvNet (CNNs) as XCOVNet for COVID-19 classification by means of chest radiograph, considering the abundance of newly developed deep learning models specifically designed for this task with more accuracy of 99.51% than previous methods.

Keywords: Radiography · Deep-Learning · Convolutional Neural Networks (ConvNet or CNN) · Artificial Intelligence (AI) · Machine Learning (ML)

1 Introduction

Since the pandemic disease (COVID-19) originating in Wuhan, China in December 2019, it has impacted billions of people. The virus caused an outbreak that spread quickly. The virus that caused COVID-19 sickness was identified after extensive investigation, and it was shown to be a member of a large family of respiratory viruses that may also cause diseases such as severe acute respiratory syndrome (SARS-CoV) and Middle East



Automated face recognition system for smart attendance application using convolutional neural networks

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Received: 4 March 2022 / Accepted: 27 November 2023
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Abstract

In this paper, a touch less automated face recognition system for smart attendance application was designed using convolutional neural network (CNN). The presented touch less smart attendance system is useful for offices and college's attendance applications with this the spread of covid-19 type viruses can be restrict. The CNN was trained with dedicated database of 1890 faces with different illumination levels and rotate angles of total 30 targeted classes. A CNN performance analysis was done with 9-layer and 11-layer with different activation functions i.e., Step, Sigmoid, Tanh, softmax, and ReLu. An 11-layer CNN with ReLu activation function offers an accuracy of 96.2% for the designed face database. The system is capable to detect multiple faces from test images using Viola Jones algorithm. Eventually, a web application was designed which helps to monitor the attendance and to generate the report.

Keywords Deep learning · Convolutional neural networks · Face database · Face recognition · Smart attendance system

1 Introduction

The present corporates companies and education organizations demands smart recognition and monitoring system with great accuracy (Gusain 2018). In the covid-19 pandemic, touch based recognition systems like thumb based biometric system may be lead to virus spread. However, touch less recognition systems helps to control the spread of covid-19 type viruses is the major motivation for the implementation. Touch less recognition system design uses either face or IRIS data (Sharma 2016). In this aspect compared

with IRIS face based recognition system will offers the good accuracy. The recognition system performance depends on the design aspects of the dataset, in that the illumination and rotation are the two basic parameters need to consider.

Among the touch less recognition options, face recognition stands out as a prominent choice. The face offers a unique and contactless means of identification, minimizing the risk of contagion. However, the implementation of a robust face recognition system is not without its challenges.

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Jacobian Based Nonlinear Algorithms for Prediction of Optimized RF MEMS Switch Dimensions

Regular Paper Published: 03 August 2023

Volume 24, pages 447–458, (2023) [Cite this article](#)[Transactions on Electrical and Electronic Materials](#)[Aims and scope](#)[Submit manuscript](#)

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Abstract

This communication discusses the role of nonlinear algorithms in training the neural network, which predicts the optimized RF MEMS switch dimensions. A dedicated dataset, i.e., *DrTLN-RF-MEMS-SWITCH-DATASET-v1*, was created by considering the most appropriate input and output variable suitable to predict the cantilever dimensions, crab leg and serpentine structure-based RF MEMS switches. The distinct artificial neural networks (ANN) performance is analysed using various training methods. The hardware implementation possible neural network algorithms, i.e., Fitting and Cascade Feed Forward Network, are considered for learning and prediction. The ANN algorithm's performance in predicting and optimizing RF MEMS switch is analysed using nonlinear



RF MEMS switch optimization using ANN and design of antenna with frequency reconfigurability

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Received: 14 May 2024 / Accepted: 15 July 2024

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Abstract

Artificial neural networks (ANN) are becoming highly prominent in the optimization of micro-RF devices, which are very significant in wireless communication applications. In this manuscript, we present the optimization of RF MEMS switches using ANN and the design of an antenna with frequency reconfigurability. A unique procedure is proposed to design reconfigurable antennas with RF MEMS switches using ANN. The novelty of this work lies in the creation of a dedicated dataset for the considered RF MEMS switch with FEM tool simulation and the utilization of cascade feed-forward neural networks for optimization. The design of the dataset and the optimization of RF MEMS switches in different aspects using ANN are the key contributions of this work. Comprehensive analysis was performed using a neural network with the designed dataset. Cascade feed-forward neural networks are highly efficient when compared with other neural networks. The weights and biases of the network were selected using the Xavier approach. The cascade feed-forward neural network is optimized using the LM training algorithm. The optimized cascade feed-forward neural network is further used to predict the optimized RF MEMS switch dimensions for the desired application. The network produces an accuracy of 94.9%. An RF MEMS switch was designed from the dimensions predicted by the cascade feed-forward neural network. The designed switch offers – 55 dB Isolation and – 0.2 dB Insertion. Eventually, an antenna was designed by incorporating identical switches which offer frequency reconfigurability.

1 Introduction

The smart wireless communication applications demand reconfigurable devices like antennas, filters, phase shifters, switches and tuners (Goel and Gupta 2020). RF MEMS technology based switches are in high demand in the design of reconfigurable communication devices i.e.,

antennas, because of their potential abilities like low power consumption and high reliability.

The materials selection, dimensions prediction and overall switch optimization are the possible research challenges in RF MEMS switches (Rikan et al. 2022). In present times, the wireless communication industry demands optimized and state-of-the-art application specific

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Robust multi-modal COVID-19 medical image registration using dense deep learning descriptor model

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<https://doi.org/10.1016/j.bspc.2024.107007> 

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Highlights

- To perform registration of multi model COVID-19 image, an enhanced residual dense learning data descriptor method is introduced.
- To remove undesirable noise from images, effective pre-processing approaches such as the Boosted Switching Bilateral Filter and the Directed Edge Enhancer algorithm for image edge enhancement are used.
- To extract intrinsic features from multi-modal COVID images, a multi-scale feature extraction module is used to reduce feature mutual interference and make training the network model easier.
- To perform the structural representation, Data Adaptive Descriptor (DAD) is utilized.
- To determine the self-similarity metrics, Sum of Squared Differences (SSD) is utilized in the Data Adaptive Descriptor.